```
"use strict";

// 1. STORAGE
const STORAGE_KEY = "personalExpenseTracker_v1";

let transactions = [];

try {
  const saved = JSON.parse(localStorage.getItem(STORAGE_KEY));
  transactions = Array.isArray(saved) ? saved : [];
} catch (error) {
  transactions = [];
}

// 2. CATEGORIES
const categories = [
  "Salary", "Food", "Transport",
  "Education", "Shopping", "Bills",
  "Health", "Entertainment", "Other"
];

// 3. SAVE DATA
function saveData() {
  localStorage.setItem(
    STORAGE_KEY,
    JSON.stringify(transactions)
  );
}

// 4. MONEY FORMAT
function money(amount) {
  return Number(amount).toLocaleString("en-IN", {
    style: "currency",
    currency: "INR",
    maximumFractionDigits: 2
```

```
  });
}

// 5. DATE
function setToday() {
  document.getElementById("date").value =
    new Date().toLocaleDateString("en-CA");
}

setToday();

// 6. TOAST MESSAGE
let toastTimer;

function showToast(message) {
  const toast = document.getElementById("toast");
  toast.textContent = message;
  toast.style.display = "block";

  clearTimeout(toastTimer);
  toastTimer = setTimeout(() => {
    toast.style.display = "none";
  }, 2500);
}

// 7. NAVIGATION
function showPage(pageId, button) {
  document.querySelectorAll(".page").forEach(page => {
    page.classList.remove("active");
  });

  document.getElementById(pageId).classList.add("active");

  document.querySelectorAll("nav button").forEach(btn => {
    btn.classList.remove("active");
  });
```

```
    button.classList.add("active");
    updateAll();
}

// 8. CALCULATE TOTALS
function calculateTotals(data = transactions) {
    const income = data
        .filter(item => item.type === "income")
        .reduce((sum, item) => sum + Number(item.amount), 0);

    const expense = data
        .filter(item => item.type === "expense")
        .reduce((sum, item) => sum + Number(item.amount), 0);

    return {
        income,
        expense,
        balance: income - expense,
        count: data.length
    };
}

// 9. UPDATE DASHBOARD
function updateDashboard() {
    const totals = calculateTotals();

    document.getElementById("totalIncome").textContent =
        money(totals.income);

    document.getElementById("totalExpense").textContent =
        money(totals.expense);

    document.getElementById("balance").textContent =
        money(totals.balance);
```

```
    document.getElementById("totalCount").textContent =
        totals.count;

    document.getElementById("calcIncome").textContent =
        money(totals.income);

    document.getElementById("calcExpense").textContent =
        money(totals.expense);

    document.getElementById("calcBalance").textContent =
        money(totals.balance);
}

// 10. ADD TRANSACTION
document.getElementById("transactionForm")
.addEventListener("submit", function(event) {

    event.preventDefault();

    const type = document.getElementById("type").value;
    const description =
        document.getElementById("description").value.trim();

    const amount = Number(
        document.getElementById("amount").value
    );

    const category = document.getElementById("category").value;
    const date = document.getElementById("date").value;

    if (!description || !Number.isFinite(amount) ||
        amount <= 0 || !date) {
        showToast("Please enter valid details!");
        return;
    }
```

```
    const transaction = {
        id: Date.now(),
        type,
        description,
        amount,
        category,
        date
    };

    transactions.unshift(transaction);

    saveData();
    updateAll();

    this.reset();
    setToday();

    showToast("Transaction added successfully!");
});

// 11. TRANSACTION HTML
function transactionHTML(item) {
    const isIncome = item.type === "income";

    const date = new Date(item.date + "T00:00:00")
        .toLocaleDateString("en-IN");

    const div = document.createElement("div");
    div.className = "transaction";

    const info = document.createElement("div");
    info.className = "transaction-info";

    const title = document.createElement("h4");
    title.textContent = item.description;
```

```
  const small = document.createElement("small");
  small.textContent = `${item.category} • ${date}`;

  info.append(title, small);

  const amount = document.createElement("span");
  amount.className = "amount " + (isIncome ? "green" : "red");
  amount.textContent =
    (isIncome ? "+" : "-") + money(item.amount);

  const deleteButton = document.createElement("button");
  deleteButton.className = "delete";
  deleteButton.textContent = "Delete";
  deleteButton.onclick = () ⇒ deleteTransaction(item.id);

  div.append(info, amount, deleteButton);

  return div;
}

// 12. DISPLAY RECENT TRANSACTIONS
function displayRecent() {
  const container = document.getElementById("recentList");
  container.replaceChildren();

  if (transactions.length === 0) {
    container.innerHTML =
      '<div class="empty">No transactions available.</div>';
    return;
  }

  transactions.slice(0, 5).forEach(item ⇒ {
    container.appendChild(transactionHTML(item));
  });
}
```

```
// 13. TRANSACTION HISTORY
function displayHistory() {
  const search = document.getElementById("search")
    .value.toLowerCase().trim();

  const filter = document.getElementById("filterType").value;

  const filtered = transactions.filter(item => {
    const matchesSearch =
      item.description.toLowerCase().includes(search) ||
      item.category.toLowerCase().includes(search);

    const matchesType =
      filter === "all" || item.type === filter;

    return matchesSearch && matchesType;
  });

  const container = document.getElementById("historyList");
  container.replaceChildren();

  if (filtered.length === 0) {
    container.innerHTML =
      '<div class="empty">No transactions found.</div>';
    return;
  }

  filtered.forEach(item => {
    container.appendChild(transactionHTML(item));
  });
}

document.getElementById("search")
  .addEventListener("input", displayHistory);

document.getElementById("filterType")
```

```
    .addEventListener("change", displayHistory);

// 14. DELETE ONE TRANSACTION
function deleteTransaction(id) {
    if (!confirm("Delete this transaction?")) return;

    transactions = transactions.filter(item ⇒ item.id !== id);

    saveData();
    updateAll();

    showToast("Transaction deleted!");
}

// 15. DELETE ALL
function deleteAll() {
    if (transactions.length === 0) {
        showToast("No transactions available!");
        return;
    }

    if (!confirm("Are you sure you want to delete all transactions?")) {
        return;
    }

    transactions = [];

    saveData();
    updateAll();

    showToast("All transactions deleted!");
}

// 16. CATEGORIES
function displayCategories() {
    const container = document.getElementById("categoryList");
```

```
  container.replaceChildren();

  categories.forEach(category => {
    const total = transactions
      .filter(item =>
        item.type === "expense" &&
        item.category === category
      )
      .reduce((sum, item) => sum + Number(item.amount), 0);

    const count = transactions.filter(item =>
      item.type === "expense" &&
      item.category === category
    ).length;

    const card = document.createElement("div");
    card.className = "category-card";

    const heading = document.createElement("h3");
    heading.textContent = category;

    const details = document.createElement("p");
    details.textContent = `${count} expense transactions`;

    const totalText = document.createElement("h2");
    totalText.textContent = money(total);

    card.append(heading, details, totalText);
    container.appendChild(card);
  });
}

// 17. MONTHLY SUMMARY
function displayMonthly() {
  const month = document.getElementById("monthSelect").value;
  const selected = document.getElementById("selectedMonth");
```

```
const allMonths = document.getElementById("allMonths");

selected.replaceChildren();
allMonths.replaceChildren();

if (month) {
  const data = transactions.filter(item ⇒
    item.date.substring(0, 7) === month
  );

  const totals = calculateTotals(data);

  const box = document.createElement("div");
  box.className = "month-card";

  const heading = document.createElement("h3");
  heading.textContent = month + " Summary";

  const income = document.createElement("p");
  income.textContent = "Income: " + money(totals.income);

  const expense = document.createElement("p");
  expense.textContent = "Expense: " + money(totals.expense);

  const balance = document.createElement("p");
  balance.textContent = "Balance: " + money(totals.balance);

  box.append(heading, income, expense, balance);
  selected.appendChild(box);
}

const months = [...new Set(
  transactions.map(item ⇒ item.date.substring(0, 7))
)].sort().reverse();

if (months.length === 0) {
```

```
    allMonths.innerHTML =
      '<div class="empty">No monthly data available.</div>';
    return;
  }

  months.forEach(monthName => {
    const data = transactions.filter(item =>
      item.date.substring(0, 7) === monthName
    );

    const totals = calculateTotals(data);

    const box = document.createElement("div");
    box.className = "month-card";

    const heading = document.createElement("h3");
    heading.textContent = monthName;

    const income = document.createElement("p");
    income.textContent = "Income: " + money(totals.income);

    const expense = document.createElement("p");
    expense.textContent = "Expense: " + money(totals.expense);

    const balance = document.createElement("p");
    balance.textContent = "Balance: " + money(totals.balance);

    box.append(heading, income, expense, balance);
    allMonths.appendChild(box);
  });
}

document.getElementById("monthSelect")
  .addEventListener("change", displayMonthly);

// 18. UPDATE EVERYTHING
```

```
function updateAll() {
    updateDashboard();
    displayRecent();
    displayHistory();
    displayCategories();
    displayMonthly();
}

// 19. INITIALIZE
updateAll();
```